



Un-fakeable Settlement

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Morpheus pays for time, not work. [verified] Settlement is `pricePerSecond` times elapsed session time (`SessionRouter.sol:264`). A bid carries one economic field, `pricePerSecond` (`IBidStorage.sol:17`); `modelId` rides along but never enters the bill. The provider's receipt signs seven fields, among them the input and output token counts (`abi.go:18-19`). The contract decodes five and drops the two counts before it settles (`SessionRouter.sol:237`). So a provider earns the same for a session whether it returned every token or none, and the same for a 7B served where a 70B was advertised. The only signature on the meter belongs to the party the payment pays. There is no slashing; the `grep` is empty. Across the 62 repos no session, usage, or revenue figure is published. Three ways to be paid for work not done, inflate volume, self-deal, or substitute a cheaper model, each cost the provider zero today.

mord settles what both sides signed. [verified repo] `mord` (github.com/luxfi/mord) is a Lux L1 that runs the marketplace as chain state, a proof of concept, built, not yet deployed or audited. It keeps three quantities apart. $\text{Capacity} = \text{Held} \times \text{PerBlock}$, the slot. $\text{Work} = (\text{In} + \text{Out}) \times \text{PerToken}$, and zero unless a meter earns it. $\text{Total} = \min(\text{Capacity} + \text{Work}, \text{Escrow})$, the rest refunded. Work pays only when the meter carries two signatures, the provider's over the token counts and the consumer's over the same counts and the output received, and only when the consumer is not the provider. A `publish` commits a model manifest digest; a `challenge` recomputes the committed model on the metered prompt, and a confirmed mismatch slashes the provider to zero and pays the challenger. Capacity settles inside block acceptance, so every validator reaches the same bill.

Faked work earns zero. [verified: 40 tests pass, 2026-08-29] Nine of the forty tests are the attacker's (`market/adversary_test.go`). A forged countersignature, a signature replayed from another session, a swapped output, a stand-in signer, and work claimed with no meter each earn zero work. A fabricated challenge slashes no one. A confirmed model substitution slashes the provider to zero stake.

Fraud surface, today vs. mord

- **Volume inflation.** Today: the provider signs its own count, and the contract that could check it discards the count. Under mord: forging volume needs a counterparty to co-sign the inflated figure.
- **Self-dealing.** Today: a provider serving itself is paid in full. Under mord: a self-dealer is paid the slot alone; work verifies to zero.
- **Model substitution.** Today: `modelId` never enters the bill, so a 7B served as a 70B pays the same. Under mord: a confirmed substitution slashes the provider to zero stake.

The three classes that cost zero today cost the attacker its whole position under mord.

What it does not prove. [assumption stated] This is bilateral attestation and operator agreement, not a zero-knowledge proof of computation. It binds who, how much, and which committed model. It does not prove the tokens were produced or the output correct; an enclave that loads the committed weights and truncates the attention window still signs a valid meter. Co-signature alone does not catch substitution; the challenge layer does. The layers compose for that reason.

Why Lux. [verified repos] Lux already runs the pieces mord composes: the attestation and slashing in `luxfi/ai` (`SlashProvider`) and the commit-reveal in `luxfi/chains/aivm`. mord is where they settle real MOR. The cheapest adoption is one layer on the diamond you already run: a meter carrying the consumer's countersignature, and a check that pays work only when both verify. No chain move, no new arithmetic; ADOPT-09 works that on-Base path end to end. We hand it over, MIT-licensed.

ADOPT-02. Companions: LP-313 verifiable inference, LP-323 separating capacity, work and identity, LP-325 linear budgets for agent coordination. Morpheus-side citations checked against the working tree; `go test ./...` in `luxfi/mord` is green. • research@lux.network